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$$\int \frac{x^2 \sqrt{x-2}}{\sqrt[3]{x-2}} dx = \int x^2 \sqrt[6]{x-2} dx$$

$$\text{Stel } u = x - 2 \Rightarrow du = dx$$

$$x^2 = x \cdot x = (u + 2)^2$$

$$= \int (u + 2)^2 \cdot u^{\frac{1}{6}} du = \int (u^2 + 4u + 4) u^{\frac{1}{6}} du$$

$$= \int u^{\frac{13}{6}} du + 4 \int u^{\frac{7}{6}} du + 4 \int u^{\frac{1}{6}} du$$

$$= \frac{6}{19} u^{\frac{19}{6}} + 4 \cdot \frac{6}{13} u^{\frac{13}{6}} + 4 \cdot \frac{6}{7} u^{\frac{7}{6}} + C$$

$$= \frac{6}{19} (x - 2)^{\frac{19}{6}} + \frac{24}{13} (x - 2)^{\frac{13}{6}} + \frac{24}{7} (x - 2)^{\frac{7}{6}} + C$$

References